Companion in the Boundary Problem network. The vehicle the Hover Disk enables · mechanism in The Warp Drive · ground-effect sibling The DIY Hoverbike · architecture How to Build a UFO · first-disk build Viktor Schauberger §9.

The Hexalock

Six hover disks in a dodecahedron — building Terrence Howard's six-pentagon collapsed-dodecahedron 6DOF airframe with a x_v hover disk in each pentagon instead of a propeller. A propellantless, moving-part-free, altitude-independent omnidirectional craft, mixed and driven by one RP2350. The vehicle the hover disk delivers the moment it first locks on the bench.

Osika Innovation AB · Alexander Osika, Claude — draft 0.3, 2026-06-20.

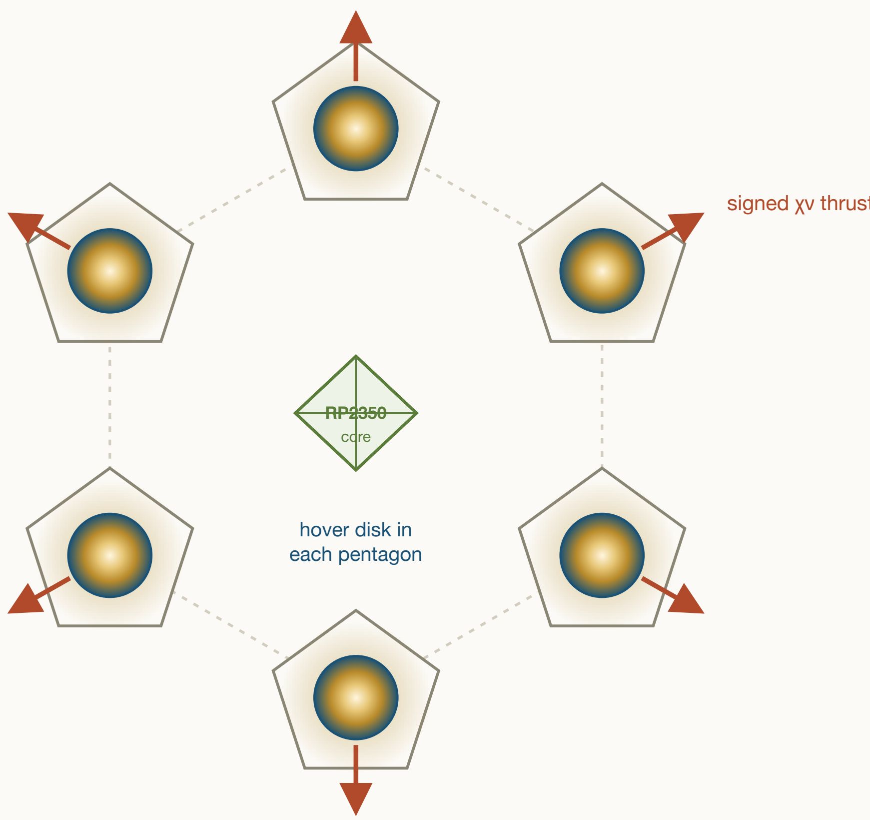

Figure 1. The Hexalock, schematic (not to scale). Six x_v hover disks — one in each pentagon of the collapsed-dodecahedron shell (dashed) — give six independent, electronically-signed thrust vectors about a central octahedral avionics + power core driven by one RP2350. No propellers; the first-article disks are the spun charged copper shells of §5.1.

> *Naming. The airframe is the published **Lynchpin** geometry; the x_v-disk craft built on it is what this paper calls the **Hexalock** — six locking disks, one lattice cell.*

*The **Lynchpin** (Howard, Molter, Seely & Yee, SAE Int. J. Aerosp. 2023) is a true six-degrees-of-freedom airframe: a collapsed dodecahedron of six pentagons, one vectored thruster per pentagon, giving six linearly independent thrust vectors and a square, invertible mixing matrix — so it translates and rotates independently ("tangential flight": hold position at any attitude, or slide any direction while level). The flown prototype put a propeller in each pentagon. **This paper puts a hover disk in each pentagon instead.** A hover disk does not lift the way a rotor does — it couples to the x_v / vertex-sharing channel, warping the spacetime lattice so the lattice itself holds the craft. To hover is to **lock**: engage the coupling and the craft is pinned to its pose the way a book is held by a table or a maglev car by its rail (`warp_drive.html` §2) — the support equals the weight, but it is a static reaction carried by the lattice, not a pumped momentum flux, so it does no work ($P = F{\cdot}v = 0$) and costs **≈ zero power**. To move is to **thrust**: modulate the coupling asymmetrically and the same disks apply a net vectored force, $P_w = (S_{coh} f_{cons}){\cdot}\kappa{\cdot}½\varepsilon_r\varepsilon_0E^2$, magnitude and sign on the drive's amplitude and asymmetry κ. That makes the disk the ideal 6DOF actuator: signed thrust with **no propeller, no spinning mass, no reversal latency, no air** — and a hover that is a lock, not a continuous lift. Six disks → six signed x_v thrust vectors → the same flight-proven 6DOF mixer, now propellantless and vacuum-capable. One **RP2350** runs it: its twelve PIO state machines generate the six disks' drive (amplitude + κ-steer) phase-locked, the two Cortex-M33 cores run the quaternion SE(3) mixer and the estimator, and the firmware is the same `12_PORT_SUBSTRATE_ADAPTER` the optical x_v chambers and the hoverbike already share. **Honest status:** this is the vehicle the hover-disk program delivers — the geometry and the 6DOF x_v mixing are sound, buildable reasoning (and the airframe's 6DOF control is flight-proven with the propeller predecessor), but the coupling itself inherits the hover disk's open status: the force law is derived in form, its magnitude is unmeasured and null to date. **This paper also documents the practically-simple first-disk build** — the Repulsine-derived spun charged copper shell of `viktor_schauberger.html` §9 (§5.1), the cheap first article for the bench, not the diamond ideal. Build the single disk that locks — self-supports its own weight — first (`hover_disk.html` bench); this is what six of them become.*

Companion papers. The thruster: `hover_disk.html` (the x_v lift body — diamond/SiC/graphene substrate ladder, the four ethers, the drive/capacitor architecture). The mechanism: `warp_drive.html` (propellantless lattice drive; §3 the force-as-pressure law, §2 why hover is ≈free, §13 the one coherence dial). Conventional sibling: `diy_hoverbike_paper.html` (the RP2350 / "engineering-optimum" template). Architecture: `how_to_build_a_ufo.html` (rim-segmented coherent-shell / saucer morphology the disk reuses). First-disk build (Architecture C): `viktor_schauberger.html` §9 (Repulsine + Tesla). Force law: `CHI_NU_FORCE_LAW.md`. Root ontology: `shape_of_the_universe.html` (the {5,3,3} dodecahedral lattice this drone is a cell of). Geometry & flight-proven 6DOF predecessor: Howard et al., SAE 2023, `doi:10.4271/01-16-03-0018`.

§1 What this is — the vehicle the hover disk enables

The opening premise of this paper is the closing promise of *The Hover Disk*: *once a single disk first locks — self-supports its own weight — on the bench, what do you build with six of them?* The answer is a Lynchpin — because the Lynchpin geometry is the one airframe that turns six independent x_v thrusters into a fully-controllable craft. Each pentagon face carries a hover disk; the six disks point along six independent axes; their x_v thrusts span the full space of forces and torques. The result is a propellantless, silent, moving-part-free, altitude-independent omnidirectional vehicle.

This is deliberately **not** the propeller drone. The flown SAE prototype used bidirectional propellers, and its own headline limitation was the latency of reversing them; the current public program moved to collective-pitch props to fight the same latency. We skip that whole branch. A hover disk has *no* rotating mass to reverse — its thrust sign is a property of the drive field's asymmetry, set in microseconds with no mechanical state — so it is not an improvement on the propeller, it is a different category of actuator. The propeller version's only role in this paper is as a de-risking *predecessor*: it proves the six-pentagon 6DOF mixing flies with real vectored thrusters (§3). We keep its flight-proven control architecture and replace the thruster.

This paper covers:

- the Lynchpin geometry and why it is also the project's spacetime cell (§2);
- the flight-proven 6DOF mixing the disks inherit (§3);
- why a x_v disk is the optimal 6DOF actuator — signed thrust, no parts, free hover (§4);
- the **practically-simple first hover-disk build** — Architecture C, the cheap spun charged copper disk to put on the bench (§5.1);
- the full RP2350 build: the six disk thrusters, the drive, the controller, docking (§5–§6);
- the field guide of what breaks, and the honest, bench-conditional status (§7, §9);
- the one coherence dial that runs this same airframe from drone to warp-craft (§8).

> **Open — to be confirmed on the bench · The load-bearing honesty, stated once, up front.** The hover disk's lift is the *open* question of the companion papers: the force law $F = (C_{sub}|x_v|^2)\cdot U_{drive}\kappa/L_{grad}$ is derived in *form*, but its *magnitude* is unmeasured and null on every substrate tested to date. This paper does not claim the disk lifts. It specifies the craft you assemble *when it does* — and shows that the geometry, the 6DOF mixing, and the RP2350 control are sound and buildable reasoning regardless, so the only open input is the one number the `warp_drive.html` §7 bench measures: where a real disk sits on the coherence dial.

§2 The geometry — a collapsed dodecahedron, which is our cell

The Lynchpin airframe is, in the SAE paper's own words, "a collapsed dodecahedron" — an icosi-dodecahedral / vector-equilibrium form built from **six pentagons** of equal edge length *a*, whose six planes are *all* mutual symmetry planes, derived from "four Tetryen elements" of Jeff Yee's energy-wave particle model. Each pentagon presents one face normal; here, each carries one hover disk, and the six normals are the craft's six thrust axes.

```
the six-pentagon "collapsed dodecahedron"          one pentagon = one hover disk
(all 6 pentagon planes are symmetry planes)        χν thruster on the face normal n̂ᵢ
                                                   signed pressure Pᵢ ∈ [−Pmax,+Pmax]
          ___________                                      n̂ᵢ
         /\         /\                                      ↑   (lattice momentum sink)
        /  \  disk /  \         6 face-normals  ⇒          ┌────────┐ rim-segmented
       / d3 \____ / d2 \        6 lin. indep.             |░░░░░░░░| diamond/SiC/graphene
       \    /\   /\    /         χν thrust vectors        |░ disk ░|  coherent χν body
        \  /  \ /  \  /         ⇒ 6×6 mixing M            |░░░░░░░░| HV/RF/THz drive,
         \/ d1 ╳ d6 \/          ⇒ full SE(3) / 6DOF        └────────┘ κ sets vector
         /\    │    /\                                     no rotor · no air · no latency
        /  \ d4│d5 /  \         "tangential flight":
       / d5 \__│__/ d4 \        translate & rotate        central octahedral
       \    /     \    /         independently            avionics + power core
        \  / four-  \  /         (orbit own CoM;          (nested Platonic solid)
         \/ bubble   \/           hover at any attitude;
            vertex                 hold ≈ free power)
        (vertex-sharing)
```

Schematic, not to scale. Six pentagons → six signed x_ν thrust vectors → a square invertible mixer → genuine six-degrees-of-freedom flight, propellantless. The vertex where pentagons meet is the "four-bubble" / four-Tetryen junction — in the companion frame, a vertex-sharing site, which is the very channel each disk locks onto.

Here is the part that is not a coincidence to anyone who has read the root papers. The whole Boundary-Problem network rests on one geometric claim: spacetime is the **{5,3,3} dodecahedral honeycomb**, whose primitive cell is the pentagon-faced dodecahedron (the carbon dodecahedrane C_{20} is its literal molecular realization; `hover_disk.html` §13). The Lynchpin's airframe *is that cell* — chosen, independently, by aerodynamicists optimizing a thrust frame — and a hover disk holds precisely *because* its body resonates with that lattice cell (the consonance ether, `warp_drive.html` §13). So putting hover disks in a Hexalock is not bolting one program onto another: it is building the lattice cell out of six smaller lattice-cell resonators. The shape that makes the best omnidirectional thrust frame is the same shape that makes the best lattice coupling.

Lynchpin term (SAE 2023 / Tangential Flight)	**Network term**	**Same object?**
collapsed dodecahedron / six pentagons	{5,3,3} primitive cell / dodecahedrane C_{20}	ESTABLISHED geometry; identification exact (same solid)
"four Tetryen elements" / four bubbles meeting	vertex-sharing site ($g \le 1/\sqrt{3}$, `G_VERTEX_DERIVATION.md`)	FRAMEWORK INFERENCE — the x_ν lift channel *is* vertex-sharing; the four-fold (tetrahedral, 109.47°) junction is a Plateau/foam vertex

Lynchpin term (SAE 2023 / Tangential Flight)	Network term	Same object?
"electromagnetic vacuum surfaces" (docking)	x_V coupling surface (the lattice momentum sink)	FRAMEWORK INFERENCE — §8; Howard's docking face *is* the disk's lift face
super-symmetric, fractal, "scales to any size"	size-independent self-lift F/W ∝ R (`hover_disk.html` §5)	DERIVED — x_V self-lift is a material property, not a size; the airframe scales the same way
central octahedral avionics core	nested Platonic solid (octahedron ⊂ dodecahedron)	ESTABLISHED — a real choice in the prototype, kept here

Doctrine sentence · *The Lynchpin airframe is the spacetime-lattice cell built as an aircraft, and the hover disk is the lattice cell built as a thruster. Assembling six disks into a Hexalock builds the cell out of cells — the same dodecahedral resonance at two scales. That is why this airframe and this thruster were made for each other: they are the same geometry, nested.*

§3 Tangential flight — six signed thrust vectors (flight-proven)

In an ordinary multirotor, translation and rotation are *coupled*: to fly forward you pitch forward, because all rotors point up. The Lynchpin's six thrusters point along six different axes, so their thrusts *span* the space of forces and torques. Collect the six unit thrust directions as the columns of a 3×6 matrix and the six moment arms likewise; stack them into the 6×6 map M from per-disk thrusts to body wrench:

$$\begin{bmatrix}\mathbf{F}\\ \mathbf{M}\end{bmatrix} = M\begin{bmatrix}P_1\\ \vdots\\ P_6\end{bmatrix}, \qquad \begin{bmatrix}P_1\\ \vdots\\ P_6\end{bmatrix} = M^{-1}\begin{bmatrix}\mathbf{F}\\ \mathbf{M}\end{bmatrix}$$

Because the six axes are linearly independent, M is square and non-singular: it **always inverts**. Name any wrench — a force in any direction, a torque about any axis, the two decoupled — and M^{-1} hands you the six thrusts that produce it (SAE 2023, Eqs. 8–10). That is the whole of "tangential flight": translate while holding attitude (stare a camera at a fixed point while orbiting it), rotate while holding position (hover upside-down, pivot in place), or both at once — the full SE(3) envelope from six thrusters.

Derived / established · **This half is flight-proven.** Howard, Molter (University of Stuttgart), Seely & Yee designed, built, and *flight-tested* the six-pentagon mixer with real vectored thrusters (a 651 g

> propeller prototype, Pixhawk 4 Mini / PX4, quaternion control), peer-reviewed in the *SAE Int. J. Aerosp.* (2023). So the geometry, the invertible mixer, and the quaternion 6DOF control law are not in question — they flew. **Swapping the thruster type from propeller to hover disk does not touch the mixer**; M^{-1} is the same matrix whether the six P_i are propeller thrusts or x_v pressures. We inherit a working control stack and change only the actuator physics.

> **Implementation consequence · The disk dissolves the prototype's one hard problem.** A pure six-thruster 6DOF vehicle must produce *signed*, arbitrarily-small thrust on each axis (the near-zero-thrust constraint that pushed ETH's Omnicopter to eight over-actuated rotors, and forced the Lynchpin prototype into bidirectional props with their reversal latency). A hover disk meets that requirement for free: its thrust is $P_w \propto \kappa \cdot U_{drive}$, continuous through zero, and its sign is the sign of the drive asymmetry κ — no spinning mass passes through zero, so there is no reversal transient at all (§4). The geometry's hardest demand is exactly the x_v actuator's natural behaviour.

One structural fact the prototype surfaced and we keep: unlike a normal multirotor, the **z-position of the centre of gravity changes the thrust solution** (the six axes are not coplanar). Keep the heavy mass — power and drive electronics — on the geometric centre in the octahedral core, and fold any residual offset into *M* at arm time, as the prototype did with its 42 mm battery offset.

§4 Lock, not lift — why a χ_v disk is the optimal 6DOF actuator

First, the word, because it changes how the whole craft reads. A propeller drone *lifts*: it runs a pump that throws air downward every second, and the upward force is the reaction to that endless momentum flux — cut the power and it falls. A x_v craft does not lift. When it holds station it **locks**: it engages the vertex-sharing coupling and the spacetime lattice *holds its pose*, the way a book is held by a table or a maglev car by its rail (`warp_drive.html` §2, the "Omicron" hover). The support still equals the weight, but it is a *static reaction carried by the lattice*, not a pumped flux, so it does no work ($P = F \cdot v = 0$) and costs ≈ zero power. To *move*, the craft modulates the coupling asymmetrically and the same disks apply a net vectored **thrust** (warp §2's "Delta" / travel) — which does cost work, recoverable regenerative-braking style. The Lynchpin has those two regimes, and "lift" names neither:

Mode	What it is	Power
Lift (PROPELLER — THE WRONG MODEL)	a continuous downward momentum pump; the reaction holds you up	continuous — stop and you fall
Lock (x_v HOVER)	engage the coupling; the lattice pins the full pose — position *and* attitude (book on a table / maglev in its rail)	**≈ 0 — no work is done; $P = F \cdot v = 0$**

Mode	What it is	Power
Thrust (χ_v MANEUVER)	modulate the coupling asymmetrically → a net force/torque to *change* the pose	costs work — recoverable (regenerative)

With that fixed, the engineering-optimum argument is not "which propeller" — it is that the propeller is the wrong category of device for this airframe, and the hover disk is the right one. Lay the two against the requirements a 6DOF craft actually has:

Requirement	Propeller (any pitch scheme)	χ_v hover disk
Signed thrust, continuous through zero	needs reversal — motor spin-down/up or pitch slew through flat	κ sign flips the gradient; smooth through zero, no transient
Reversal / response latency	the prototype's stated bottleneck (tens of ms / servo-limited)	field-set; microseconds, no mechanical state to change
Moving parts per thruster	rotor, bearings, (servo, linkage)	none in the materials-ideal disk; one constant-RPM flywheel in the simple Architecture-C first build (§5.1) — never a thrust-reversing part
Works without air	no — thrust → 0 with density; ceiling ~30 km	yes — the hold is a lattice lock, altitude-independent
Power to *lock* a hover	continuous (100s of W) — pumping against gravity every second	≈ 0 — the lattice holds the pose; P = F·v = 0 (warp §2)
Acoustic signature	broadband rotor noise	silent (no aerodynamic source)
Thrust direction within a face	fixed along the shaft	steerable — rim-segment κ can vector within the disk (bonus authority)
Coupling magnitude — THE CATCH	known, available today	OPEN — null to date; the bench question (§9)

The disk wins every row but the last, and the last is the whole game. The disk's thrust is the warp pressure of `warp_drive.html` §3, written per face:

$$P_{\mathrm{w}} = \left(S_{\mathrm{coh}}\, f_{\mathrm{cons}}\right) \cdot \kappa \cdot \tfrac{1}{2}\varepsilon_r \varepsilon_0 E^2, \qquad \text{lock condition} \quad P_{\mathrm{w}} \geq \rho\, t\, g$$

with S_{coh} the coherent fraction (life ether), f_{cons} the consonance fidelity (chemical ether, ceiling $e^{-P/12} \approx 0.873$), $\kappa \in [-1,1]$ the drive asymmetry that sets the vector's *magnitude and sign*, and $\tfrac{1}{2}\varepsilon_r\varepsilon_0 E^2$

the stored field-energy density (≈ 25 MPa for a diamond stack at 1 GV/m). A disk reaches the lock condition — supports its own weight — when its drive pressure exceeds its areal weight $\rho t g$ — a *size-independent* condition (the area cancels), so it is a property of the disk's *material and drive*, not its diameter. The warp paper's §13 dial says the coupling $S_{coh}f_{cons}$ need only clear $\sim 10^{-6}$ for a thin diamond skin to carry its own weight — a sliver of full coherence.

> **Implementation consequence · Thrust is a number you write to a register.** Each disk's commanded thrust P_i maps to a drive setpoint (|amplitude| → field E, sign → κ). There is no rotor to govern, no pitch to slew, no thermal/aero lag — the actuator is the drive waveform. That is why the §3 mixer becomes trivial to close: M^{-1}[**F**;**M**] lands on six clean, fast, linear, bipolar actuators with a true zero. The hardest control problem in 6DOF flight — authority at zero net thrust — is the disk's easy regime.

> Doctrine sentence · *A propeller converts shaft power into a momentum flux of air, every second, downward; a hover disk borrows a force from the spacetime lattice and gives it back when it lets go. The first is a pump that must run to hold you up; the second is a foot that stands on the floor. For a craft that must hover at any attitude, in any medium, the disk is not a better pump — it is the end of pumping. The craft does not lift; it* ***locks*** *to hover and* ***thrusts*** *to move.*

§5 The build — six disks, one RP2350, twelve channels

5.1 The disk thruster — start practically simple (Architecture C)

`hover_disk.html` ranks the *best* disk: single-crystal diamond in its optical mode, an icosahedral-quasicrystal register, the full four-ether stack. That is the destination, not the first article. The drone's first disks should be a "**pretty simple**" **weekend build**, in the hoverbike's spirit — the cheapest body that exercises enough of the levers to be worth putting on the `warp_drive.html` §7 force balance. The companion `viktor_schauberger.html` §9 names exactly that build, reasoned from history rather than materials science: **Architecture C — the spun, charged, chiral-vortex copper shell.** It is the first-disk spec we adopt, and the reason this drone paper, not the hover-disk paper, is where the simple build lives: the hover-disk paper optimizes the *material*; here we optimize the *first thing you actually machine*.

Schauberger's Repulsine, with Tesla's bladeless turbine and resonant coil beside it, assembles *four* of the warp drive's actuation levers (`warp_drive.html` §4) in copper, with shop tools and no cleanroom: **spin** (a mechanical κ-axis symmetry-breaker), **stored HV charge** (U_{drive}), a **chiral-spiral field** (the asymmetry κ built into the geometry, not the electronics), and a **closed, cold shell** (coherence). It needs no THz/optical drive and no exotic crystal — a spun, charged copper disk does it. The framework's own honesty argues this is the *right* first try: the published electrostatic-thrust nulls (`hover_disk.html` §15) are all *static* capacitors with no high-frequency mode (no predicate P4), so they cannot even test the x_v channel — and bulk spin supplies that mode *mechanically*, cheaply, the way no static plate can.

First-article layer	Practically-simple pick (per pentagon)	Lever it buys (warp §4)
Body / shell	two facing domed copper discs across a corrugated "waveline" gap, ~80–120 mm (the pentagon's inscribed disc)	closed cold shell — the Repulsine geometry
Spin	one small balanced BLDC spinning the shell at a governed constant RPM (a flywheel, *not* a propeller)	spin = the κ-axis symmetry-breaker (warp §4-i)
Charge / electrode	a chiral-spiral electrode etched on the disc faces, charged to a few kV	stores U_{drive}; the spiral *is* κ's direction, no steering electronics
HV stage	a small per-disk flyback or Cockcroft–Walton, RP2350-gated; optional Tesla-coil-style resonant arm	delivers/stores U_{drive}; the resonant arm adds the P4 high-frequency field
Coupling skirt (optional)	a smooth or textured boundary-layer disc (the Tesla turbine)	a *mechanical* η coupling channel
Cooling	passive heatsink / thermal mass; colder is strictly better	the coherence budget — Schauberger's "cold = living"

The §3–§4 control contract is met with no exotic part. Per disk the RP2350 commands **thrust magnitude** via the HV charge amplitude (electronic, fast) and **thrust sign / vector** via charge polarity / differential bias across the spiral (electronic); the spin is held at a constant RPM by a slow governor and is a *symmetry-breaker, not a control*. So the mixer still writes {amplitude, sign} to six disks (§5.2) — with the one honest caveat that the simple first disk carries a constant-RPM flywheel (a balanced rotor, no thrust reversal, far below the complexity of the variable-pitch propeller it replaces).

> **Open — to be confirmed on the bench · This is a first bench article, not a working thruster.** Build *one*, put it on the `warp_drive.html` §7 vacuum force balance, and measure self-lift — against the stored-energy, κ→0, and chamber-variation controls — *before* you build six and bolt them to an airframe. Architecture C is the open build hypothesis of `viktor_schauberger.html` §9: it exercises four levers the static capacitor does not, which is why it is the right first try — not a guarantee that it lifts. Every x_v measurement to date is null; this is the cheapest configuration that could change that.

The upgrade ladder — when the simple disk returns a receipt

The moment a simple disk shows a non-null force on the balance, climb the `hover_disk.html` material ladder for more coupling per disk, and eventually to the no-moving-parts static endpoint (which trades the spin for a high-frequency optical drive). Same socket in the pentagon, better fill:

Disk layer	Materials-ideal pick	Why (which ether)
Structural / coherent core	single-crystal diamond (^{12}C-enriched); SiC for a cheaper larger article	energy ceiling (Warmth, R = 1084) + coherence (Life); optical-mode η in one body
Geometry register	face etched as a 12-fold photonic quasicrystal	consonance (Chemical) — the icosahedral/φ lattice match, in a true insulator
Electrodes	graphene, rim-segmented (≈12 segments)	atomically smooth (raises breakdown); the segments are the κ-steer (replacing the spin)
Coupling option	thin $BiFeO_3$ (hBN-buffered)	coupling (Light) — the multiferroic P2 channel
Drive	HV with an RF→THz arm (per-segment phase/amplitude)	stores U_{drive}; exercises the §4 A/B frequency fork — and removes the moving part

5.2 Why the RP2350, and the channel map

We keep the project's standard controller for the same reasons the hoverbike does, plus one specific to this airframe:

- **Twelve PIO state machines.** The RP2350 has three PIO blocks × four state machines = **12** cycle-exact I/O engines. Six disks, each driven by two channels (amplitude gate + κ-segment phase), is **12 channels** — one PIO state machine each, generating the disk's HV/RF drive timing with no jitter and no CPU in the loop. The fit is one-to-one.
- **Two M33 cores for the two loops.** Core 0 runs the quaternion SE(3) mixer (the M^{-1} solve + frame transforms, already quaternion-native in the SAE control scheme) and the per-disk thrust→drive map; Core 1 runs the attitude/rate controller, the EKF (IMU/baro/flow), and the disk health/coherence monitor.
- **Firmware reuse — and it is the same machine.** The output stage is the `12_PORT_SUBSTRATE_ADAPTER` the optical x_v chambers use. The optical chamber *is* a 12-port x_v drive; a Hexalock is six x_v drives on a flying frame. The coordinator, phase scorebook, and resonance-locking warm loop transfer unchanged; below the adapter the per-port stage drives a disk's HV gate instead of a laser current.
- **Determinism & cost.** A ~$1 MCU owns the timing in hardware (PIO), not on a contended RTOS.

```
RP2350   ┌──────────────────────── Core 0 ─────────────────────────┐
         │ 6DOF mixer:  P[1..6] = M⁻¹ · [Fx Fy Fz Mx My Mz]ᵀ   (quaternion) │
         │ per-disk map: P_i → (|E|_i drive amplitude, κ_i asymmetry/sign) │
         └──────────────────────────────────────────────────────────┘
         ┌──────────────────────── Core 1 ─────────────────────────┐
         │ EKF (IMU+baro+optical-flow) → attitude q, rates ω, pos; outer │
         │ attitude/rate PID → wrench [F;M]; coherence/HV health; failsafe │
         └──────────────────────────────────────────────────────────┘
  PIO0  SM0..3  →  disk 1,2,3,4  drive-amplitude gates (to HV/RF stage)
  PIO1  SM0..1  →  disk 5,6      drive-amplitude gates
        SM2..3  →  disk 1,2      κ-steer: rim-segment phase, or spiral-bias (Arch. C)
  PIO2  SM0..3  →  disk 3,4,5,6  κ-steer: rim-segment phase, or spiral-bias + spin-sync
  HW    I2C/SPI →  IMU (ICM-42688-P), baro (BMP390), mag, flow/ToF
        UART    →  RX (ELRS/CRSF), telemetry / inter-craft mesh
        ADC/PWM →  per-disk field/HV sense; docking-face drivers; LED/buzzer
```

One MCU owns all twelve drive channels deterministically. PIO is the low-voltage timing layer; the kilovolt drive lives in a per-disk HV/RF stage (the PIO drives its gate), exactly as the hoverbike's PIO drives class-D amps. Six disks × {amplitude, κ} = the 12 PIO state machines.

5.3 Airframe

Keep the prototype's proven structure. A **central octahedral core** (3D-printed centerpiece + six corners, tied by 1.5 mm CFRP triangles) carries the RP2350 board, the power core, and the IMU at the geometric centre; the **outer dodecahedral shell** is CFRP tube and printed pentagon connectors, with the six disks seated on the six pentagon faces. The octahedron-in-dodecahedron is a real packaging win (a rigid central pyramid network) and a nested Platonic solid — the avionics live in the dual of the shell. The disks need a clear, undisturbed face to the outside (their coupling surface), so unlike a ducted rotor there is nothing to shroud and no airflow to manage — the shell can be closed and smooth, which is also what the coherence story of §8 wants.

5.4 Sensing & control

The estimator is conventional (IMU + baro + optical-flow/ToF indoors; GNSS + mag outdoors). The human interface follows the hoverbike's "one input device" doctrine, because a 6DOF craft has no inherent "up" to fall back on:

- **Autonomous / pose-hold (default).** The craft holds a pose in SE(3) at ≈zero power (§4); the operator commands waypoints and look-directions. Six-DOF station-keeping is an autopilot capability, not a stick skill — and here it is also nearly free to sustain.
- **Two-stick decoupled.** Left stick = body translation (**F**), right stick = body rotation (**M**) — literally the two halves of M^{-1}.
- **VR / first-person.** The SAE paper's own recommendation: head-tracked goggles with the thrust axes aligned to the pilot's view, so "push forward" is always "where I'm looking." The RP2350 needs only the head quaternion on a UART.

5.5 Docking & swarm — the coupling faces

The Lynchpin's symmetric pentagon faces let units **dock in flight** and act as one: the SAE design uses face connectors with spring-dampers (soft capture) and pogo pins (a drone-to-drone bus for comms and power), and a *master-aircraft* scheme where, on connection, one craft solves the others' thrust via a dynamically enlarged mixing matrix (N docked Hexalocks → one 6N-thrust vehicle). In our frame the docking faces and the lift faces are the *same surface* — the x_v coupling face (§8) — so bonding two craft is also widening a coherent boundary. Four units close into a gripper (Config A) or a sphere that catches a payload (Config B); that sphere is, not incidentally, a step toward the closed coherent shell of §8.

§6 Mass, power, and the endurance inversion

A Hexalock's budget is unlike a multirotor's, because nothing converts power into a continuous downward air-flux. Numbers are ESTIMATES for sizing, conditional on the disk (§9); they assume a thin diamond/SiC-class disk at the §13 self-lift coupling.

Component	Mass	Notes
6× hover disks (thin diamond/SiC core + graphene rim + register)	~6×60 = 360 g	self-lift is per-weight & size-independent ($F/W \propto R$); thinner is easier, not harder
6× per-disk HV/RF drive stages	~6×35 = 210 g	stores U_{drive}; the heaviest electronics on the craft
Octahedral core + dodeca shell (CFRP)	~150 g	closed, smooth — no ducts, no airflow paths
RP2350 board + wiring + sensors	~40 g	whole flight computer in a palm
Power core (battery / supercap; or fuel cell / rectenna option)	~150 g	sized for *maneuver work* + *losses*, not for hover
Docking faces + connectors (swarm units)	~60 g	optional
Total (solo)	**~850–950 g**	comparable to the propeller version — but the power story is inverted

For the **Architecture-C first build** (§5.1), swap the diamond skins for copper shells and add six small spin motors + drivers (~6×25 = 150 g): the solo craft lands ~1.0–1.1 kg. The materials-ideal disks later remove the motors and that mass with them. Either way the budget is dominated by *drive electronics*, not by lift hardware — there is no continuous-lift power plant to carry, because the craft *locks* rather than pumps.

Derived / established · **The endurance inversion — lock, don't lift (warp §2).** Holding a hover does no work — the craft **locks**, the lattice holding it the way a table holds a book (P = F·v = 0), so a pose-hold draws only the drive field's *losses* (a low-loss / superconducting-boundary disk approaches free hover). Travel costs work, but a conservative coupling is *regenerative*: energy spent accelerating out is recovered decelerating in (KERS-style), so a round trip costs only losses plus net work — not the gross momentum changes. So unlike a multirotor that burns its whole pack just to stay up, a Hexalock's battery is sized for *maneuvering*, and station-keeping endurance is set by losses, not by lift power. *(This is the lawful claim; "free energy / never runs down" is not — you recover what you put in, minus losses.* `warp_drive.html` *§2.)*

§7 Things that will go wrong

Field guide, in the hoverbike's spirit — the failure modes of a moving-part-free x_v 6DOF craft:

1. **The disk may not lift at all — that is the headline risk.** Every x_v measurement to date is null. Build and fly the *single self-lifting disk* on the `warp_drive.html` §7 force balance before you assemble six into an airframe. This vehicle is downstream of that receipt; do not invert the order.
2. **κ-steer calibration is the new mixer input.** The map from commanded thrust P_i to (amplitude, κ) per disk must be characterized per disk on a balance, because the §3 mixer assumes a clean, signed, linear actuator. A nonlinear or hysteretic κ→force curve re-introduces the very problem collective pitch was trying to remove — table-ize it.
3. **HV breakdown and field health.** The disks store kilovolt fields; breakdown, leakage, or aging shift U_{drive} and thus thrust. The Core-1 health monitor must watch each disk's field and fail a disk safely (the §3 mixer can fly degraded on five, with reduced authority).
4. **The simple disk spins — balance it.** Architecture C's constant-RPM flywheel must be balanced (vibration decoheres the shell and corrupts the IMU), and its gyroscopic moment folded into the attitude model; six co-rotating discs add net angular momentum, so alternate spin directions across the six (or null it in the estimator). The materials-ideal static disk has none of this — the spin is the price of the cheap first build, and the reason to climb off it once the bench gives a receipt.
5. **Coherence loss looks like thrust loss.** S_{coh} (the coherent fraction) is temperature- and drive-sensitive; a disk that decoheres weakens silently. Treat thermal management of the coherent core as flight-critical, the way a multirotor treats motor temperature.
6. **CG-z still bites.** A vertical CG offset couples into the non-coplanar thrust solution (§3). Keep mass centered in the octahedral core; re-calibrate *M* after any payload change.
7. **Disorientation before the craft.** A vehicle with no inherent "up" is disorienting line-of-sight. Fly pose-hold or VR first; two-stick manual is an expert mode.

8. **Docking is a contact-dynamics problem.** Two 6DOF craft capturing face-to-face is where the spring-dampers and a soft-capture state machine earn their mass; dock slow, with the master pre-chosen, and bench the matrix-handover before flying it.

§8 One airframe, one dial — drone to warp-craft

Because the thruster is already a x_v drive, this airframe does not need an "upgrade path" to reach the warp program — it *is* the warp program's first vehicle, sitting at the low end of the one coherence dial of `warp_drive.html` §13. The same six (→twelve) faces, driven harder and more coherently, walk the same shell up the ladder:

Rung	Coupling $S_{coh}f_{cons}$	What the Hexalock is	Status
lock (hover)	$\sim 1.4\times 10^{-6}$	this drone — six partly-coherent disks, hover & gentle flight, 0.1–1 g	OPEN (the bench result this paper is downstream of)
flight envelope	$\sim 10^{-3}$	the Tic-Tac envelope — partial coherent shell, ~700 g, fast 6DOF	OPEN — same disks, stronger drive
closed shell	$\to 0.873$	the full dodecahedron driven as one coherent shell (the "expanded" Hexalock; or a Config-B swarm sphere) — near-c, then the warp *closes*	HIGH-GATE — warp §8 (FTL bubble)

Two design facts of the Hexalock make it unusually well-placed on this dial, by virtue of being this shape: it is a *closed, maximally-symmetric dodecahedral shell* (the wholeness/life ether S_{coh} wants a closed body; the consonance/chemical ether f_{cons} wants exactly the {5,3,3} cell), and its faces are already *independently driven with controllable asymmetry* (the κ the force direction needs). The "collapsed" six-pentagon form is the drone; the full twelve-face dodecahedron — or four craft bonded into a Config-B sphere — is the closed coherent shell of the starship limit. Howard's swarm bonding is, in this frame, a coherence-area multiplier: docked shells share a wider boundary, raising S_{coh} by aggregation.

> Doctrine sentence · *One airframe, one dial. Six hover disks in a dodecahedron is a drone at one part in a million of full coherence; the same dodecahedron brought to whole-shell coherence is a warp-craft. The Hexalock does not graduate from drone to starship by changing shape — it climbs the coherence-and-consonance curve of the warp drive while staying the exact same cell. Build the drone; you have built the first rung of the starship.*

§9 What is established, and what is open

Element	Status
Six-pentagon collapsed-dodecahedron geometry; six lin.-indep. thrust vectors; M square & invertible → 6DOF	ESTABLISHED (SAE 2023, derived & flight-tested)
The 6DOF mixer / quaternion control flies with real vectored thrusters	ESTABLISHED — the propeller predecessor flew; the mixer is thruster-agnostic
A x_v disk gives signed, latency-free, moving-part-free, air-free thrust, and ≈free hover	DERIVED from the force law & warp §2 — *conditional* on the disk lifting at all
RP2350: 12 PIO SMs = 6 disks × {amplitude, κ}; dual-M33 mixer/estimator; firmware reuse from the 12-port x_v adapter	ESTABLISHED capability; the mapping is THIS PAPER'S DESIGN
Airframe = the {5,3,3} lattice cell; disks lift by resonating with that cell	ESTABLISHED geometry; the lattice identification is the network ontology
The hover disk's lift magnitude — does a real disk self-lift?	OPEN — force law derived in *form*; magnitude unmeasured, null to date. The whole vehicle is downstream of this one number.
The flight / Tic-Tac envelope and the closed-shell warp limit	OPEN / HIGH-GATE — `warp_drive.html` §8

> **Open — to be confirmed on the bench ·** The honest one line: **this is the vehicle the hover disk delivers, not a vehicle that flies today.** The geometry, the flight-proven 6DOF mixing, and the RP2350 control are sound and buildable now; the lift is the single open bench question of the companion papers. The right order is the `hover_disk.html` tier ladder — first the self-lifting disk on the `warp_drive.html` §7 force balance, then six of them in this airframe. We never let the Lynchpin's flight-proven control half lend its certainty to the x_v lift half; they are two ledgers, and only one is closed.

§10 Conclusion

Put a propeller in each pentagon and the airframe is a clever omnidirectional drone. Put a hover disk in each pentagon and it is something else: a propellantless, silent, moving-part-free craft that hovers at any attitude in any medium for almost no power, mixed and driven by a single RP2350 whose twelve channels fall naturally onto the six disks' drive. The airframe's 6DOF control is flight-proven; the disk's lift is the open bench question of the hover/warp papers — and the contribution of this paper is to show that the geometry, the mixing, and the controller are ready and waiting, so that the day a single disk self-lifts, six of them are a craft. It is the same dodecahedral cell all the way up the coherence dial: a drone at a millionth of full coherence, a starship at the whole shell. Build the self-lifting disk first. Then build six, and fly the cell.

References

1. Howard, T.D., Molter, C., Seely, C.D. & Yee, J. (2023). "The Lynchpin — A Novel Geometry for Modular, Tangential, Omnidirectional Flight." *SAE Int. J. Aerosp.* 16(3), Article 01-16-03-0018, `doi:10.4271/01-16-03-0018`. (Geometry, the six-thrust-vector mixing matrix, the flight-tested 6DOF prototype and quaternion control, the docking/swarm configurations. Used here for the geometry and the flight-proven control; the propeller propulsion is replaced.)
2. *The Hover Disk*, `hover_disk.html` — the x_v lift body that fills each pentagon: the substrate ladder (diamond/SiC/graphene), the four ethers, the rim-segmented drive and capacitor architecture, and the honest derived/open status the disk's lift carries.
3. *Viktor Schauberger — The Water Wizard and the Inward Drive*, `viktor_schauberger.html` §9 — "Architecture C, the spun charged chiral-vortex shell" (Repulsine + Tesla bladeless turbine + resonant coil): the practically-simple first-disk build of §5.1, a bench hypothesis exercising four warp-drive levers the static capacitor does not. Right shape; the lift is the open bench question.
4. *The Warp Drive*, `warp_drive.html` — the propellantless lattice-drive mechanism; §3 the force-as-pressure / self-lift law, §2 why hover is ≈free and travel is regenerative, §13 the closed-form thrust and the one coherence dial (self-lift $\sim 10^{-6} \leftrightarrow$ FTL bubble).
5. *The DIY Hoverbike*, `diy_hoverbike_paper.html` — the RP2350 / 12-port substrate adapter / "one input device" engineering-optimum template this build follows.
6. *How to Build a UFO*, `how_to_build_a_ufo.html` §6/§10 — the closed-shell, rim-segmented saucer morphology each disk and the whole shell reuse.
7. *The Shape of the Universe*, `shape_of_the_universe.html`; *The Periodic Table*, `the_periodic_table.html` §13 — the {5,3,3} dodecahedral lattice and the dodecahedrane C_{20} primitive cell the airframe instantiates. Vertex-sharing bound: `G_VERTEX_DERIVATION.md` ($g \leq 1/\sqrt{3}$); force law: `CHI_NU_FORCE_LAW.md`.
8. Brescianini, D. & D'Andrea, R. — the ETH "Omnicopter," an eight-rotor 6DOF vehicle; Ryll, M. (2015), overactuated quadrotor (University of Stuttgart). (The 6-thruster near-zero-thrust constraint of §3, which the x_v disk dissolves.)
9. Raspberry Pi Ltd. — *RP2350 Datasheet* (dual Cortex-M33 + dual RISC-V Hazard3; three PIO blocks / twelve state machines). The §5 controller.

Draft engineering paper in the x_v / vertex-sharing research program. It specifies a propellantless 6DOF craft that places a hover disk (the x_v lift body of `hover_disk.html`) in each pentagon of the flight-proven Lynchpin geometry (Howard et al., SAE 2023). The geometry and the 6DOF mixing/control are established and flight-proven with a propeller predecessor; the **lift itself is the open bench question** of the companion papers — the force law is derived in *form*, its magnitude unmeasured and null to date — and the whole vehicle is downstream of that single measurement (`warp_drive.html` §7). Quantities are tagged ESTABLISHED / DERIVED / ESTIMATE / OPEN. The Lynchpin geometry and "tangential flight" are trademarks of Tangential Flight Corporation; this paper analyzes and builds on the published design and claims no affiliation.